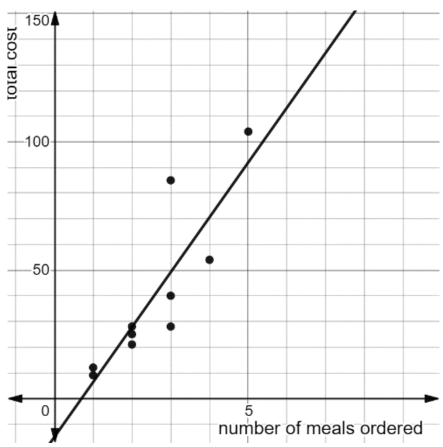
Algebra 1
Unit 3
Lesson H1 Homework

Name____Answer key______
Date____________________
Period__________________

A restaurant took a random sample of customers ordering takeout meals. The data from the sample is shown.

Number of Meals ordered	1	2	5	3	2
Total Cost	\$12	\$21	\$104	\$28	\$25
Number of Meals ordered	4	3	3	1	2
Total Cost	\$54	\$85	\$40	\$9	\$28

This data is represented by a scatter plot with the function $c(x) = 21.42x - 15.08$ fit to the data, where c represents the total cost of the order and x represents the number of meals ordered.



For questions 1 through 5, complete the tables to find the residual values for each number of meals ordered.

1.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
1	\$12	\$ 6.34	5.66
1	\$9	\$ 6.34	2.66

2.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
2	\$21	\$ 27.76	-6.76
2	\$25	\$ 27.76	-2.76
2	\$28	\$ 27.76	0.24

3.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
3	$28	$ 49.18	−21.18
3	$40	$ 49.18	−9.18
3	$85	$ 49.18	35.82

4.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
4	$54	$ 70.60	−16.6

5.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
5	$104	$ 92.02	11.98

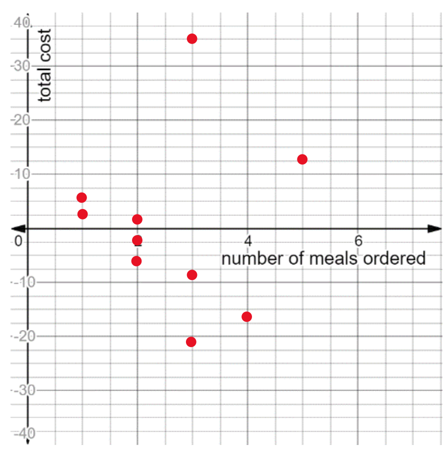
6. How many positive residuals does the data set have?

5

7. What is the residual with the largest absolute value?

35.82 (3 meals with actual cost $85)

8. Plot the residuals on the coordinate grid.



9. Explain whether the linear function is a good fit for this data set. Answers may vary

The residual values are quite spread out & fairly large, so I do not think the linear function is a good fit for this data set.